Supplementary Material for TRACE: Context-Probed Region Allocation for Test-Time Adaptation in Open-Vocabulary Semantic Segmentation

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A Exact Method and Evaluation Protocol

A.1 Episode, Image, Crop, and Candidate

The production method is `ImageBalancedTRACE` in mode C. It contains no second probe, semantic-role branch, candidate deletion, dominance attenuation, fragment guard, or CLS loss. An episode resets pretrained parameters, Adam state, and the prescribed RNG state; candidates and diagnostics are then constructed once, followed by updates and final inference on unmodified inputs. Only executed visual LayerNorm affine parameters train. Text encoding, projection weights, and other backbone parameters stay frozen.

A source image is the normalization unit, a crop is an encoder input, and a token is a spatial encoder output. The general loader uses 224×224 sliding windows with stride 112 and row-major positions; proposals from overlapping windows remain distinct. The evaluated configuration first resizes each input to 224×224, giving one crop per source. ViT-L/14 has a 16×16 token grid; its patch size 14 is not a crop size. The valid-token mask excludes padding, not annotated background.

Four-connected components of reset-model query argmax define (b_R, Ω_R, c_R). Components are not merged across query extensions even if they map to one evaluation category. Source-image top-K uses descending mean query probability, descending token area, ascending query index, and ascending component index; stable crop/component enumeration resolves remaining exact ties. There is no additional area threshold beyond nonemptiness, stitching, or overlap union. This confidence-selected pool is shared by the matched allocation controls, not a test of confidence-independent admission.

The deterministic runtime uses PyTorch deterministic algorithms with errors on nondeterminism, `CUBLAS_WORKSPACE_CONFIG=:4096:8`, deterministic cuDNN, and disabled cuDNN benchmarking. Default attention dispatch remains enabled. Seed 0 and original full-evaluation orders are fixed. These execution choices do not create independent replications.

Actual first-input gates compare the specified direct/worker computations with fixed 10^{-5} numerical tolerance, zero changed labels, and finite loss, gradient, parameter, optimizer, and final-support checks as applicable. That agreement tolerance is not the loss's 10^{-8} EPS threshold. Component gates do not certify semantic correctness, full published-pipeline equivalence, or independent scientific replication. They precede full execution; separate CPU scoring receives annotation permission only after the required complete prediction seals and actual successful exit.

A.2 Context Transform and Score

The C transform denormalizes with channel means $(122.7709, 116.7460, 104.0937)$ and standard deviations $(68.5005, 66.6322, 70.3232)$ to RGB values in $[0, 255]$. Its contrast coefficient is 0.35. Per-channel means for the transform are computed over the entire current crop, including candidate pixels; transformed channels shift (r, g, b) to (g, b, r), then clip. Nearest-neighbor upsampling of the original token mask determines preserved pixels. The original predicted query is read on the original tokens after re-normalization and reset-model inference. Supports are means of query probabilities, not logits, ratios, or CLS scores. Candidates are not re-extracted from transformed predictions.

Each selected candidate requires one diagnostic input. Views are stacked in bounded batches and their outputs mapped back to their original candidates. Diagnostic encoder calls therefore depend on batching, while input count remains one per candidate. Scores are detached and fixed through the episode. Preservation is geometric, not an intervention that isolates causal evidence: full-crop statistics and contextual feature mixing remain. The compatibility name `counterfactual_support` does not certify a causal counterfactual.

A.3 Two Dense Readouts

Diagnosis and final inference aggregate the last 18 selected visual outputs of reduced NACLIP ViT-L/14 (one-based blocks 7–24) with crop-specific entropy weights. Neighborhood standard deviation is 5. Selected outputs come from the attention branch,

while earlier residual/MLP paths propagate normally. The reduced final block omits its unused MLP. Optimization instead averages selected layers uniformly and uses separate prompt embeddings. Neither readout is interchangeable with averaging probabilities across layers.

Let $h_{\ell,b,i}$ be the selected pre-projection output and $\psi_\theta(h) = \mathrm{LN}_{\mathrm{post},\theta}(h)W_{\mathrm{proj}}$. A hat denotes unit normalization, $t_{a,c}$ is a normalized frozen text embedding, and $\bar{t}_c = \widehat{J^{-1}\sum_a t_{a,c}}$. The frozen pretrained logit multiplier is $\lambda = \exp(\ell_{\mathrm{CLIP}})$. Inference computes

$$d_{\ell,b,i}(c) = \lambda\langle\widehat{\psi_\theta(h_{\ell,b,i})}, \bar{t}_c\rangle, \tag{1}$$

$$v_{\ell,b} = N^{-1}\sum_i H(\mathrm{softmax}_c\, d_{\ell,b,i}), \tag{2}$$

$$w_{\ell,b} = \frac{\exp(-v_{\ell,b})}{\sum_m \exp(-v_{m,b})}, \tag{3}$$

$$z^{\mathrm{eval}}_{b,i}(c) = \lambda\left\langle \psi_\theta(\widehat{\sum_\ell w_{\ell,b}h_{\ell,b,i}}), \bar{t}_c \right\rangle. \tag{4}$$

Here $N = 256$ spatial tokens and CLS is excluded. Diagnosis uses θ_0; final inference uses adapted parameters. Optimization uses the gradient-bearing logits

$$z^{a,\mathrm{adapt}}_{b,i}(c) = \lambda\left\langle \psi_\theta(\widehat{L^{-1}\sum_\ell h_{\ell,b,i}}), t_{a,c} \right\rangle, \tag{5}$$

where $L = 18$ in the primary L/14 setting. The seven fixed templates are:

```
itap of a {}
a bad photo of the {}.
a origami {}.
a photo of the large {}.
a {} in a video game.
art of the {}.
a photo of the small {}.
```

Prompt embeddings and text parameters never train. The official OpenAI CLIP ViT-L/14 checkpoint SHA-256 is the concatenation

```
b8cca3fd41ae0c99ba7e8951adf17d26
7cdb84cd88be6f7c2e0eca1737a03836.
```

The B/16 generalization conditions use the declared official B/16 checkpoint, last 9 of 12 outputs, and their pinned G preprocessing. SCLIP's final MLP is executed, so the legitimate trained-LayerNorm membership differs from reduced NACLIP; matching an incorrectly identical parameter count would implement a different comparison.

A.4 Objective and Reconstruction

Region entropy averages over prompts and tokens before within-image weighting. The active test is $\rho_R > 10^{-8}$. The helper converts detached weights to the entropy dtype and tests the summed denominator in double precision, using one as a fallback denominator if it does not exceed EPS. With no active source, the graph-connected zero is returned and the engine skips adaptation. Adam has $\beta_1 = 0.9$, $\beta_2 = 0.999$, zero weight decay, and the frozen dataset-specific learning rate. Visual input/output and executed-block LayerNorm affine parameters train; the unused final-MLP normalization in reduced NACLIP remains frozen. No global image/CLS objective is added.

For finite fixed active sets, $\sum_{R\in\mathcal{R}_j^+}\alpha_{R|j} = 1$. Thus each active source has coefficient mass $1/|\mathcal{I}_+|$, whereas episode normalization assigns it a mass proportional to its total support. Common positive rescaling within an image cancels only while the active set stays unchanged; a single active candidate has coefficient one before the outer average. A token's explicit coefficient is $\alpha_{R|j}/(J|\mathcal{I}_+||\Omega_R|)$. Region averaging removes a direct area multiplier, but not area-dependent features, overlapping proposals, or shared-parameter effects. Joint Adam is not a sum of independent candidate Adam updates, and equal source budgets do not imply batch invariance.

Final query logits are bilinearly upsampled with `align_corners=False`, placed at crop offsets, and averaged over overlap coverage. The general loader does not force a terminal crop; uncovered positions have zero logits. The present one-crop square evaluation covers every evaluated pixel. Pixelwise query softmax precedes taking the maximum over extension queries for each evaluation category, followed by category argmax. This is not a sum of extension probabilities or category collapsing before candidate extraction.

A.5 Population and Metric

VOC20/VOC21 and PC59/PC60 respectively share source images but differ in label protocols. The numbers of classes with positive pooled union are 20, 21, 59, 60, 81, 171, 3, and 3 for VOC20, VOC21, PC59, PC60, Object, Stuff, CD, and Waterbirds. Object contains 80 foreground classes plus background. VOC20/PC59 exclude background, whereas VOC21/PC60/Object/CD/Waterbirds include it. Ignore value 255 is excluded. The reported metric is

$$\mathrm{mIoU} = \frac{100}{|\mathcal{Y}_+|} \sum_{y \in \mathcal{Y}_+} \frac{I_y}{U_y}, \quad \mathcal{Y}_+ = \{y : U_y > 0\}, \tag{6}$$

with intersections/unions pooled over the declared whole population. It is not mean image IoU, candidate IoU, or a binary foreground/thing-role IoU.

Native M02/M03/M04/V01 RGB uses PIL bilinear and historical PIL-nearest label resize at 224×224. G/G03 instead use the declared OpenCV RGB geometry and palette-label decoding with OpenCV nearest at that resolution. Different resize tie conventions can change labels; none is substituted for another in a comparison. Query vocabularies/extensions, source IDs/order, grouping, checkpoints, and readouts remain pinned within each study.

A.6 Consumed Dataset and Corruption Protocols

Datasets and labels. We consume existing image/mask pairs, not newly constructed splits. VOC20/21 share the VOC2012 validation list (Everingham et al. 2015); PC59/60 share the local Pascal-Context validation list (Mottaghi et al. 2014) under the configured VOC2012-layout root. Object uses COCO val2017 (Lin et al. 2014); Stuff uses the local COCO-Stuff label protocol (Caesar, Uijlings, and Ferrari 2018). The table specifies retained local populations, not a claim to reproduce every public release or its original preprocessing. Ordered class-name files and exact source/config pins accompany this text.

Protocol	Sources	Labels	Consumed mask/split convention
VOC20	1,449	20	VOC validation; original background 0 ignored; foreground IDs reduced by 1.
VOC21	1,449	21	Same images/masks; background 0 retained.
PC59	5,105	59	Context validation; original background 0 ignored; foreground IDs reduced by 1.
PC60	5,105	60	Same images/masks; background 0 retained.
Object	5,000	81	val2017; existing `instanceTrainIds` masks; 80 objects plus background.
Stuff	5,000	171	val2017; existing `labelTrainIds` masks; ordered thing/stuff labels, no extra background class.
COCO-CD	500	3	Existing `test_balanced` IDs/masks; 0 background, 1 cat, 2 dog.
Waterbirds-Seg	5,322	3	All original-loader pairs; 0 background, 1 landbird, 2 waterbird.

The native loader converts images to RGB, resizes images by PIL bilinear and masks by PIL nearest-neighbor to 224×224, and uses 224×224 patches with stride 112. Value 255 is ignored; only VOC20/PC59 additionally turn original background 0 into ignore and reduce the remaining valid IDs by 1. Other protocols retain existing IDs. Same-relative-stem image/mask pairs are sorted by image pathname, including when selected by a split list.

COCO-CD: the local COCO cat/dog protocol. COCO-CD here names the consumed COCO-derived cat/dog segmentation protocol, not an additional official COCO task. Within `coco_cd_ovss`, `splits/test_balanced.txt` selects 500 JPEGs from `images/` and same-stem PNGs from `masks/test_balanced/`. The loader neither generates masks nor remaps their labels. The retained audit records a separate 1,668-source declared training population (`splits/train_corr0.95.txt`, `masks/train_corr0.95/`); its 256-source calibration panel has zero own-evaluation source-ID/path/inode overlap, not certified global historical independence. *Unrecovered construction:* the historical 500-ID selection/balancing recipe, annotation rasterization, category-ID mapping, instance-overlap rule, and any image/background compositing were not recovered from the bounded sources checked. The word *balanced* is the retained split name, not a new group-balance verification. These are consumed existing masks, not an independently reconstructed benchmark.

Waterbirds-Seg: segmentation pairs and split exposure. The public Waterbirds generator (Sagawa et al. 2020) composites CUB bird foregrounds with Places backgrounds, distinguishes landbird/waterbird and land/water backgrounds, and sets training correlation to 0.95. This upstream classification recipe does not establish the creation of the local three-class masks or the exact local subset. The consumed root is `waterbirds_seg/correlation_0.95`. That directory name is not a fresh verification of local group frequencies. Its frozen full loader has no split-list filter: it pairs all existing JPEGs under `images/` with same-relative-stem PNGs under `masks/`, including species subdirectories, retaining 5,322 pairs.

Let F denote this unchanged full development population, T the declared training sources from `splits/train.txt`, and C the selected calibration panel. The retained audit establishes

$$C \subset T \subset F, \qquad |F| = 5{,}322, \quad |T| = |T \cap F| = 2{,}458, \quad |C| = |C \cap F| = 256.$$

Thus `train_full_source_id_overlap`=2,458 and `cal_full_source_id_overlap`=256 are different nested-set relationships, not additive counts; 256 does not describe all training/full overlap. Neither count asserts that CLIP or TRACE was trained on 2,458 images. Calibration ranks training IDs by SHA-256 of the UTF-8 string formed by `trace-single-probe-tuning-v1`, seed 2027, the dataset key, and official ID, each followed by a newline; ties use ID, the first 256 are selected, then sorted by original source pathname. An older selected panel also has 512/512 sources overlapping F. Historical development exposure remains; this is not an independent test. *Unrecovered construction:* the local CUB/Places pairing, mask-writing landbird/waterbird assignment, crop/resize, foreground threshold, compositing RNG, and original 5,322-pair subset construction were not recovered. The audit records no new construction. We specify the consumed mask protocol, not an independently reproducible reconstruction of its creation.

Corruption implementation and parameters. VOC20-C uses the frozen MLMP-derived imagecorruptions reference (Noori et al. 2025), recorded at commit `2a887de018b3f2c3b132f69fe806bab12a2d511f`. This identifies the local frozen implementation, not an unspecified current package or universal ImageNet-C equivalence. Recorded versions are NumPy 1.26.4, SciPy 1.15.3, scikit-image 0.25.2, Pillow 12.2.0, and OpenCV 4.11.0. The severity-indexed reference parameters actually used are below; dashes denote unrequested combinations.

Corruption	$s = 1$	$s = 3$	$s = 5$
Gaussian noise	0.08	0.18	0.38
Shot noise	—	—	3
Impulse noise	—	—	0.27
Defocus blur	—	—	$(10, 0.5)$
Glass blur	—	—	$(1.5, 4, 2)$
Motion blur	$(10, 3)$	$(15, 8)$	$(20, 15)$
Zoom blur	—	—	$\mathrm{arange}(1, 1.31, 0.03)$
Snow	—	—	$(0.55, 0.3, 2.5, 0.85, 12, 12, 0.55)$
Frost	—	—	$(0.6, 0.75)$
Fog	$(1.5, 2)$	$(2.5, 1.7)$	$(3, 1.4)$
Brightness	—	—	0.5
Contrast	0.4	0.2	0.05
Elastic transform	—	—	$\alpha = 30$
Pixelate	—	—	0.25
JPEG compression	25	15	7

Parameters mean: Gaussian standard deviation on the unit scale; Poisson scale for shot noise; per-channel impulse amount with salt fraction 0.5; defocus (radius, alias blur); glass (Gaussian sigma, maximum shift, iterations); motion (radius, sigma), angle uniform on $[-45, 45)$ degrees; zoom factors from the exact NumPy `arange` call; snow (mean, standard deviation, zoom, threshold, motion radius, motion sigma, image-blend weight); frost (image, frost) blend weights; fog (fractal amplitude, decay); brightness HSV-value increment; contrast multiplier about channel means; pixelate downscale ratio (BOX downsampling, nearest upsampling); JPEG quality. Elastic transform uses $\sigma = 0.01(H, W)$, severity-5 $\alpha = 30$, both pre-smoothing displacement ranges $[-0.005H, 0.005H)$, and order-1 reflect interpolation. All function lines and parameter expressions are pinned in provenance. The six packaged assets are `frost/frost1.png`–`frost3.png` and `frost4.jpg`–`frost6.jpg`; their declared hashes are retained. The unchanged `randint(5)` selector samples only the first five.

For each of the 1,449 ordered original VOC sources, each requested type/severity seeds NumPy with the zero-based original source index and restores its previous RNG state afterward. OpenCV-decoded BGR data enter the pinned reference, including its PIL conversion, without an intervening BGR-to-RGB correction. Corruption precedes OpenCV `INTER_LINEAR` resize to 224×224; lossless PNGs are consumed without a second image resize. The 15 severity-5 types give 21,735 inputs. The five curve types (Gaussian noise, motion blur, fog, contrast, JPEG) add severities 1 and 3, giving 14,490 inputs; severity 5 reuses sealed G1954 predictions/counts, not new inference. Labels are not warped by the generator, including for elastic corruption. G/G03 scoring uses palette indices with OpenCV `INTER_NEAREST` to 224, distinct from native PIL-nearest scoring. G03 analysis was registered after known G02 severity-5 outcomes and before new severity-1/3 predictions. These are historically exposed fixed-scope diagnostics, not newly untouched generalization evidence. This specification changes no population, score, split, mask, or experiment.

Table 1: TRACE-selected configurations: episode size B, learning rate, update steps, and maximum regions per image K. Uniform and Confidence reuse these settings without separate tuning. Selection uses 256 calibration images per dataset; the Waterbirds calibration sources overlap its full development population. Historical development exposure is retained.

Dataset	B	Learning rate	Steps	K
VOC20	2	0.0015	9	8
VOC21	18	0.00125	31	46
PC59	6	0.002	10	46
PC60	13	0.00375	7	20
Object	13	0.00375	30	35
Stuff	7	0.00425	10	33
COCO-CD	26	0.015	5	4
Waterbirds	12	0.005	4	7

B Selection and Exposure

One global mode C was selected on the original Object/PC60 training-source cohorts, before eight-dataset parameter search. A label-free hash ranking with seed 20261029 selected 128 calibration images and 256 distinct confirmation images per dataset after documented exclusions. The selection maximized the worst-dataset C/T/minimum margin over Confidence, using a 0.02-pp tie band, then fewer diagnostic inputs, larger mean margin, and fixed mode order. C's PC60 calibration margin was negative (-0.272957 pp) and Object's was $+0.579561$ pp. T and minimum had worse worst-dataset margins. Confirmation did not reselect the mode.

On the fixed 256-image confirmation cohorts, C exceeded Confidence across the original order and two permutations by $+0.069026$ to $+0.176377$ pp on PC60 and $+2.100626$ to $+2.924691$ on Object. The legacy minimum was better than C on all three PC60 orders, and T was better on all Object orders. Reordered cohorts reuse images and are not independent samples. Historical July 4,096-image candidate-pool membership cannot be exhaustively certified from surviving loader records; the exclusions do not establish never-seen data.

Parameter search then fixed C, geometry, prompts, backbone, and readouts and varied batch size, learning rate, steps, and K on seed-2027 256-image training-source panels. Coordinate coarse/fine search evaluated 401 configurations for seven datasets and 54 for Waterbirds. Selection maximized calibration mIoU; within 0.02 pp, lower model time and a fixed configuration tuple broke ties. The cap was 80 unique configurations per dataset. Full scores did not trigger further search, and Uniform/Confidence were not separately tuned. The selected configurations are retained verbatim in Table 1.

Historical development and test exposure remains even though predictions precede scoring. In particular, Waterbirds' 256 calibration images are members of its unchanged 5,322-source evaluation population. The deterministic eight-protocol comparison therefore describes TRACE-selected operating points with development overlap on Waterbirds, not unseen evaluation or independently tuned baselines. Later studies use fixed inherited settings; their bootstrap intervals do not include uncertainty from that selection process.

Table 2: M01 inside-region evidence. Unadjusted and fixed-matching estimates retain all signed outcomes at threshold 0.001. Harm differences are percentage points; Support is a difference in mean true-class probability change, not percentage points or mIoU. Matching intervals are null when required strata lack a denominator; U counts undefined draws. Coverage is high/low-quintile counts for unadjusted rows and matched/total candidates plus stratum count S for matched rows.

Dataset	Step	Contrast	Endpoint	Estimate [95% CI]	U / draws	Coverage
Object	1	C Q5 - Q1	Harm	-13.659 [-23.994, -3.566]	0/10000	205/205
Object	1	C Q5 - Q1	Support	-0.010 [-0.049, +0.033]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Harm	-2.439 [-12.698, +7.626]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Support	-0.083 [-0.126, -0.042]	0/10000	205/205
Object	1	Score matched	Harm	-9.412 [null]	10000/10000	494/1024; S=190
Object	1	Score matched	Support	+0.011 [null]	10000/10000	494/1024; S=190
Object	1	Score + GT quality	Harm	-1.708 [null]	10000/10000	298/1024; S=120
Object	1	Score + GT quality	Support	+0.013 [null]	10000/10000	298/1024; S=120
Object	30	C Q5 - Q1	Harm	-11.707 [-21.330, -2.168]	0/10000	205/205
Object	30	C Q5 - Q1	Support	-0.053 [-0.112, +0.008]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Harm	+2.927 [-6.352, +11.878]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Support	-0.267 [-0.329, -0.204]	0/10000	205/205
Object	30	Score matched	Harm	-5.022 [null]	10000/10000	494/1024; S=190
Object	30	Score matched	Support	-0.003 [null]	10000/10000	494/1024; S=190
Object	30	Score + GT quality	Harm	+1.667 [null]	10000/10000	298/1024; S=120
Object	30	Score + GT quality	Support	-0.020 [null]	10000/10000	298/1024; S=120
PC60	1	C Q5 - Q1	Harm	-26.471 [-36.182, -16.553]	0/10000	204/204
PC60	1	C Q5 - Q1	Support	+0.021 [-0.014, +0.055]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Harm	-18.137 [-29.972, -6.331]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Support	-0.024 [-0.065, +0.017]	0/10000	204/204
PC60	1	Score matched	Harm	-2.065 [null]	10000/10000	626/1018; S=192
PC60	1	Score matched	Support	-0.018 [null]	10000/10000	626/1018; S=192
PC60	1	Score + GT quality	Harm	+6.020 [null]	10000/10000	434/1018; S=161
PC60	1	Score + GT quality	Support	-0.033 [null]	10000/10000	434/1018; S=161
PC60	7	C Q5 - Q1	Harm	-28.431 [-38.650, -18.214]	0/10000	204/204
PC60	7	C Q5 - Q1	Support	+0.067 [+0.025, +0.108]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Harm	-25.490 [-35.249, -15.277]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Support	+0.027 [-0.042, +0.092]	0/10000	204/204
PC60	7	Score matched	Harm	-3.426 [null]	10000/10000	626/1018; S=192
PC60	7	Score matched	Support	+0.001 [null]	10000/10000	626/1018; S=192
PC60	7	Score + GT quality	Harm	+4.881 [null]	10000/10000	434/1018; S=161
PC60	7	Score + GT quality	Support	-0.022 [null]	10000/10000	434/1018; S=161

C Candidate-Risk Panel

M01 selects 128 sources per dataset with a stable hash and samples up to eight candidates without their outcome. The resulting 1,024 Object and 1,018 PC60 candidates come from the existing confidence-top-K pools, in 113 and 105 original source episodes. Each candidate starts from reset model, optimizer, and RNG, and adapts with unit weight. This is not its coefficient within a joint TRACE update. Readouts occur at initial, step 1, and final (30 Object / 7 PC60) states; no midpoint was recorded or invented.

The primary harm event is a drop exceeding 0.001 in mean true-class probability inside the original candidate. Frozen quintiles compare high and low C; matching additionally controls confidence width 0.05, class and original query, and token-area bins, then initial GT-correctness quartile in a second analysis. The estimator equally averages fixed-stratum high-minus-low-C contrasts, not a pooled candidate effect. Ground-truth quality is offline-only and is not a candidate selection signal or deployed upper bound.

Score-only matching covers 494/1,024 Object candidates in 190 strata and 626/1,018 PC60 candidates in 192 strata. Adding initial GT quality covers 298 candidates in 120 strata and 434 in 161. Both matching analyses at both checkpoints/datasets have 10,000/10,000 undefined draws and null intervals. Ties and unmatched coverage need not form disjoint categories. Conditioning reverses the final point differences, leaving independent risk unestablished, not proving a harmful causal effect. The canonical sealed result JSON retains inside/outside/whole regions and thresholds 0, 0.001, and 0.005; none is reselected for a positive conclusion.

Table 3: M01 outside-region evidence. Definitions, units, frozen threshold, null handling, and coverage columns match Table 2; these secondary outcomes are not substituted for the primary inside-region estimand.

Dataset	Step	Contrast	Endpoint	Estimate [95% CI]	U / draws	Coverage
Object	1	C Q5 - Q1	Harm	-10.244 [-19.929, -0.218]	0/10000	205/205
Object	1	C Q5 - Q1	Support	+0.017 [+0.008, +0.025]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Harm	-4.878 [-13.844, +3.951]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Support	+0.003 [-0.004, +0.011]	0/10000	205/205
Object	1	Score matched	Harm	-6.776 [null]	10000/10000	494/1024; S=190
Object	1	Score matched	Support	+0.007 [null]	10000/10000	494/1024; S=190
Object	1	Score + GT quality	Harm	-6.528 [null]	10000/10000	298/1024; S=120
Object	1	Score + GT quality	Support	+0.004 [null]	10000/10000	298/1024; S=120
Object	30	C Q5 - Q1	Harm	-8.780 [-18.750, +1.123]	0/10000	205/205
Object	30	C Q5 - Q1	Support	+0.052 [+0.004, +0.101]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Harm	+5.366 [-4.937, +15.650]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Support	-0.063 [-0.112, -0.012]	0/10000	205/205
Object	30	Score matched	Harm	-2.588 [null]	10000/10000	494/1024; S=190
Object	30	Score matched	Support	+0.016 [null]	10000/10000	494/1024; S=190
Object	30	Score + GT quality	Harm	+3.611 [null]	10000/10000	298/1024; S=120
Object	30	Score + GT quality	Support	+0.001 [null]	10000/10000	298/1024; S=120
PC60	1	C Q5 - Q1	Harm	-12.745 [-22.789, -2.599]	0/10000	204/204
PC60	1	C Q5 - Q1	Support	+0.016 [+0.006, +0.026]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Harm	-10.784 [-22.472, +0.634]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Support	+0.016 [+0.004, +0.028]	0/10000	204/204
PC60	1	Score matched	Harm	-0.141 [null]	10000/10000	626/1018; S=192
PC60	1	Score matched	Support	+0.001 [null]	10000/10000	626/1018; S=192
PC60	1	Score + GT quality	Harm	+4.555 [null]	10000/10000	434/1018; S=161
PC60	1	Score + GT quality	Support	-0.002 [null]	10000/10000	434/1018; S=161
PC60	7	C Q5 - Q1	Harm	-14.216 [-23.407, -5.013]	0/10000	204/204
PC60	7	C Q5 - Q1	Support	+0.040 [+0.013, +0.066]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Harm	-22.059 [-34.271, -9.356]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Support	+0.074 [+0.029, +0.124]	0/10000	204/204
PC60	7	Score matched	Harm	-0.727 [null]	10000/10000	626/1018; S=192
PC60	7	Score matched	Support	+0.005 [null]	10000/10000	626/1018; S=192
PC60	7	Score + GT quality	Harm	+0.983 [null]	10000/10000	434/1018; S=161
PC60	7	Score + GT quality	Support	-0.004 [null]	10000/10000	434/1018; S=161

Table 4: M01 whole-image evidence, with the same definitions as Table 2. Both saved checkpoints and all matched null intervals remain; no midpoint or precision from dropped draws is supplied.

Dataset	Step	Contrast	Endpoint	Estimate [95% CI]	U / draws	Coverage
Object	1	C Q5 - Q1	Harm	-10.732 [-20.333, -0.792]	0/10000	205/205
Object	1	C Q5 - Q1	Support	+0.017 [+0.009, +0.026]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Harm	-5.366 [-14.034, +3.214]	0/10000	205/205
Object	1	Confidence Q5 - Q1	Support	+0.003 [-0.005, +0.010]	0/10000	205/205
Object	1	Score matched	Harm	-6.645 [null]	10000/10000	494/1024; S=190
Object	1	Score matched	Support	+0.007 [null]	10000/10000	494/1024; S=190
Object	1	Score + GT quality	Harm	-6.319 [null]	10000/10000	298/1024; S=120
Object	1	Score + GT quality	Support	+0.003 [null]	10000/10000	298/1024; S=120
Object	30	C Q5 - Q1	Harm	-9.268 [-19.407, +0.902]	0/10000	205/205
Object	30	C Q5 - Q1	Support	+0.050 [+0.002, +0.099]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Harm	+4.390 [-5.825, +14.606]	0/10000	205/205
Object	30	Confidence Q5 - Q1	Support	-0.064 [-0.113, -0.013]	0/10000	205/205
Object	30	Score matched	Harm	-2.061 [null]	10000/10000	494/1024; S=190
Object	30	Score matched	Support	+0.015 [null]	10000/10000	494/1024; S=190
Object	30	Score + GT quality	Harm	+4.444 [null]	10000/10000	298/1024; S=120
Object	30	Score + GT quality	Support	-1.37e-04 [null]	10000/10000	298/1024; S=120
PC60	1	C Q5 - Q1	Harm	-14.706 [-24.455, -4.926]	0/10000	204/204
PC60	1	C Q5 - Q1	Support	+0.018 [+0.008, +0.028]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Harm	-10.784 [-22.217, +0.509]	0/10000	204/204
PC60	1	Confidence Q5 - Q1	Support	+0.015 [+0.003, +0.028]	0/10000	204/204
PC60	1	Score matched	Harm	+1.552 [null]	10000/10000	626/1018; S=192
PC60	1	Score matched	Support	+3.50e-04 [null]	10000/10000	626/1018; S=192
PC60	1	Score + GT quality	Harm	+6.108 [null]	10000/10000	434/1018; S=161
PC60	1	Score + GT quality	Support	-0.003 [null]	10000/10000	434/1018; S=161
PC60	7	C Q5 - Q1	Harm	-15.196 [-24.628, -5.920]	0/10000	204/204
PC60	7	C Q5 - Q1	Support	+0.045 [+0.019, +0.071]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Harm	-22.549 [-34.773, -9.953]	0/10000	204/204
PC60	7	Confidence Q5 - Q1	Support	+0.073 [+0.028, +0.123]	0/10000	204/204
PC60	7	Score matched	Harm	-0.987 [null]	10000/10000	626/1018; S=192
PC60	7	Score matched	Support	+0.005 [null]	10000/10000	626/1018; S=192
PC60	7	Score + GT quality	Harm	+2.226 [null]	10000/10000	434/1018; S=161
PC60	7	Score + GT quality	Support	-0.004 [null]	10000/10000	434/1018; S=161

Table 5: M02: all four mode scores and three prespecified paired contrasts for each full population. Arm values are pooled mIoU; contrast values are percentage points. All intervals are pointwise 95%, with 10,000 whole original-episode draws and zero undefined draws.

Dataset	Mode / contrast	Estimate [95% CI]	U / draws
Object	Native	32.671 [31.786, 33.291]	0/10000
Object	Translated	31.843 [30.931, 32.461]	0/10000
Object	Global	31.644 [30.733, 32.270]	0/10000
Object	Confidence	29.361 [28.552, 30.016]	0/10000
Object	Native - translated	+0.828 [+0.433, +1.236]	0/10000
Object	Native - global	+1.027 [+0.605, +1.446]	0/10000
Object	Native - Confidence	+3.310 [+2.646, +3.881]	0/10000
PC60	Native	30.783 [30.263, 31.230]	0/10000
PC60	Translated	30.678 [30.173, 31.120]	0/10000
PC60	Global	30.666 [30.163, 31.106]	0/10000
PC60	Confidence	30.573 [30.076, 30.998]	0/10000
PC60	Native - translated	+0.105 [-0.029, +0.234]	0/10000
PC60	Native - global	+0.117 [-0.015, +0.245]	0/10000
PC60	Native - Confidence	+0.210 [+0.064, +0.351]	0/10000

D Spatial, Legacy, and Factorial Controls

D.1 Spatial Preservation

M02 holds the original candidate, readout tokens, loss locations, and configurations fixed. Translated preservation has equal mask area but changes its position/content. The original 16×16 mask is first upsampled to input pixels; each axis supplies its minimum legal offset, zero, and maximum legal offset. Among their Cartesian-product translations excluding zero displacement, the rule minimizes overlap, then maximizes squared displacement, then minimizes lexical (dy, dx). If no displacement exists, the unchanged mask is retained as a fallback. Translation never clips, wraps, rotates, or resizes the mask. The global condition preserves no candidate pixels and deduplicates its diagnostic input per source crop. All modes still read the original query at original candidate tokens. These contrasts test specified pipeline operators rather than isolated candidate causal utility.

Object contains 173,503 candidates, with native/translated/global active counts 171,161/172,885/172,898; Confidence activates all. Translation has 173,043 shiftable and 460 fallback masks. PC60 has 101,505 candidates, corresponding active counts 101,002/101,179/101,179, and 100,891/614 shiftable/fallback. Natural EPS differences remain in the estimand. Native-minus-translated/global Object intervals are positive; both PC60 spatial intervals cross zero (Table 5).

D.2 Declared Legacy Components and Objective

M03 crosses C versus a declared archived diagnostic with image versus episode normalization. The archive contributes five modules (constants, role taxonomy, reliability, indexing, and interventions) only. Its role-aware diagnostic schedules three views for thing candidates and two for stuff/background; C schedules one per candidate. Roles enter that declared comparator, not production C or the shared role-free confidence-top-K pool. The current engine, Adam, readout, and resource bounds remain. It is not a whole historical pipeline, old-engine/optimizer reproduction, or OOM-retry restoration. The archived source SHA-256 is

```
d88873fa15ada2751e07d7129cb30e96c
94f832fbf4c053d63b10384f6b1dc46.
```

Executed legacy views and score (M03). The declared legacy comparator embeds archived constants, roles, reliability, indexing, and intervention components into the current engine; it is not a whole historical-method reproduction. Let $R = (b_R, \Omega_R, c_R)$ be a region from the unchanged confidence-top-K pool, and let M_R be its original token mask upsampled to input pixels by nearest-neighbor interpolation. For normalized input x, write $u = x \odot s_0 + m_0$ for RGB pixels, where $m_0 = (122.7709, 116.7460, 104.0937)$ and $s_0 = (68.5005, 66.6322, 70.3232)$; $N(v) = (v - m_0)/s_0$ channelwise. The two fixed context replacements are

$$
\begin{aligned}
g(u) &= 0.2989u_r + 0.5870u_g + 0.1140u_b, \\
B(u) &= \mathrm{Rep}_3\big(\mathrm{AvgPool}_{9,1,4}(g(u))\big), \qquad (7)\\
A(u) &= \mathrm{clip}_{[0,255]}\big(\mathrm{Shift}(\bar{u} + 0.35(u - \bar{u}))\big), \qquad (8)
\end{aligned}
$$

where the pool has kernel 9, stride 1, zero padding 4, and divisor 81 also at borders; Rep_3 repeats the gray channel, $\bar{u}$ is the full-crop mean of each channel, and $\mathrm{Shift}(r, g, b) = (g, b, r)$. No stochastic transform is added. The target-preserved, residual-context, and context-perturbed views are

$$
\begin{aligned}
x_R^T &= N\big(M_R \odot u + (1 - M_R) \odot B(u)\big), \\
x_R^R &= N\big((1 - M_R) \odot u + M_R \odot B(u)\big), \qquad (9)\\
x_R^C &= N\big(M_R \odot u + (1 - M_R) \odot A(u)\big).
\end{aligned}
$$

The implementation mixes the corresponding normalized tensors; these equations give its real-arithmetic definition. At reset parameters, every measured view uses the same diagnostic readout, original query c_R, and original tokens:

$$
s_R^v = \frac{1}{|\Omega_R|} \sum_{i \in \Omega_R} \pi^0_{b_R,i}(c_R \mid x_R^v), \qquad v \in \{T, R, C\}. \tag{10}
$$

Thing candidates schedule (T, R, C), in that order; stuff and background candidates schedule (T, C) and do not measure s_R^R. Supports are clipped to $[0, 1]$ before the archived scalar rule:

$$
\begin{aligned}
q_R &= \min(s_R^T, s_R^C), \\
d_R &= \begin{cases} \max(s_R^R - s_R^T, 0), & \text{thing}, \\ 0, & \text{stuff/background}, \end{cases} \qquad (11)\\
\rho_R^{\text{legacy}} &= \mathrm{clip}_{[0,1]}\big(q_R(1 - d_R)\big). \qquad (12)
\end{aligned}
$$

Initial confidence selects candidates but is not an additional factor in this formula. Production single-C instead uses $\rho_R^C = s_R^C$, one view per candidate, with no role or residual penalty. Neither branch extracts new regions or switches queries after transformation.

Exact role inheritance. Roles are assigned to benchmark categories first, then inherited by queries. Category names are stripped and lowercased. For Object, the literal category `background` is background and every other category is thing. For PC60, `background` is background; the following literal names are thing:

aeroplane, bag, bed, bench, bicycle, bird, boat, book, bottle, bus, car, cat, chair, computer, cow, cup, dog, horse, keyboard, motorbike, mouse, person, plate, pottedplant, sheep, sofa, table, train, truck, tvmonitor.

Every other PC60 category is stuff. If query q maps to evaluation category $m(q)$, its role is $r(m(q))$, not a classification of the query's spelling. With class extensions, comma-separated vocabulary groups are flattened in category/group order and every query in group k maps to k; without extensions the map is identity. Background renaming applies to the benchmark category name before role assignment. Thus query extensions retain separate candidate/query identities while inheriting the same category role. Roles do not alter the shared role-free candidate pool or production-C diagnostic.

Table 6: M03: all four diagnostic/objective modes and five paired contrasts per dataset. Arm scores are pooled mIoU; contrasts are percentage points with pointwise 95% intervals. The opposite primary directions are retained.

Dataset	Mode / contrast	Estimate [95% CI]	U / draws
Object	C / image	32.671 [31.769, 33.290]	0/10000
Object	C / episode	32.624 [31.756, 33.246]	0/10000
Object	Legacy / image	33.051 [32.173, 33.685]	0/10000
Object	Legacy / episode	33.008 [32.145, 33.645]	0/10000
Object	C image - legacy image	-0.380 [-0.742, -0.040]	0/10000
Object	C episode - legacy episode	-0.384 [-0.744, -0.027]	0/10000
Object	C: image - episode	+0.047 [-0.227, +0.278]	0/10000
Object	Legacy: image - episode	+0.044 [-0.255, +0.299]	0/10000
Object	Diagnostic-objective interaction	+0.003 [-0.327, +0.333]	0/10000
PC60	C / image	30.783 [30.270, 31.218]	0/10000
PC60	C / episode	30.852 [30.335, 31.288]	0/10000
PC60	Legacy / image	30.641 [30.125, 31.080]	0/10000
PC60	Legacy / episode	30.715 [30.196, 31.153]	0/10000
PC60	C image - legacy image	+0.142 [+0.032, +0.250]	0/10000
PC60	C episode - legacy episode	+0.137 [+0.030, +0.242]	0/10000
PC60	C: image - episode	-0.069 [-0.134, -0.006]	0/10000
PC60	Legacy: image - episode	-0.074 [-0.138, -0.006]	0/10000
PC60	Diagnostic-objective interaction	+0.005 [-0.078, +0.084]	0/10000

The executed diagnostic-by-normalization crossing. For either score $s \in \{C, \text{legacy}\}$, set $\mathcal{R}_j^{+,s} = \{R \in \mathcal{R}_j : \rho_R^s > 10^{-8}\}$ and $\mathcal{I}_+^s = \{j : \mathcal{R}_j^{+,s} \neq \varnothing\}$. Using the same mean regional prompt entropy $\overline{H}_R(\theta)$ on *unmodified* inputs,

$$\overline{H}_R(\theta) = \frac{1}{J|\Omega_R|} \sum_{a=1}^{J} \sum_{i \in \Omega_R} \left[-\sum_q p^a_{\theta, b_R, i}(q) \log p^a_{\theta, b_R, i}(q) \right], \tag{13}$$

the two objectives are

$$\mathcal{L}^s_{\text{image}}(\theta) = \frac{1}{|\mathcal{I}_+^s|} \sum_{j \in \mathcal{I}_+^s} \frac{\sum_{R \in \mathcal{R}_j^{+,s}} \text{sg}(\rho_R^s) \overline{H}_R(\theta)}{\sum_{R \in \mathcal{R}_j^{+,s}} \text{sg}(\rho_R^s)}, \tag{14}$$

$$\mathcal{L}^s_{\text{episode}}(\theta) = \frac{\sum_j \sum_{R \in \mathcal{R}_j^{+,s}} \text{sg}(\rho_R^s) \overline{H}_R(\theta)}{\sum_j \sum_{R \in \mathcal{R}_j^{+,s}} \text{sg}(\rho_R^s)}. \tag{15}$$

Empty regions and weights $\leq 10^{-8}$ are omitted; if no term is active, updates are skipped. Weights and regions are measured once at reset, detached, and held fixed during the episode. These are regional means, not area-weighted pixel sums or query pseudo-label cross-entropy. In episode normalization an image's total coefficient is proportional to its retained reliability sum; image normalization gives each active image mass $1/|\mathcal{I}_+^s|$. The four modes are C/image, C/episode, legacy/image, and legacy/episode. The receipt alias `episode_min` selects episode normalization only; it does not substitute a minimum-support diagnostic for the chosen score.

Full modes and all five contrasts appear in Table 6. Image-objective C loses on Object and wins on PC60, with the same signs under episode normalization. Object's two image-minus-episode intervals cross zero; PC60's are negative. Both diagnostic-by-objective interactions cross zero, not an equivalence test. Diagnostic input count and measured raw time remain descriptive costs, not an equal-time efficacy result.

D.3 Distribution by Ordering

Executed assignment and support order (M04). Assignments are independent *within each original source image*, never across images or the whole episode. Both raw scores κ_R and C_R are measured on the same original selected candidates before any assignment. For natural-active use the entire selected set $S_j = \mathcal{R}_j$. For the separate common-active study first restrict to

$$S_j = \{R \in \mathcal{R}_j : \min(\kappa_R, C_R) > 10^{-8}\}. \tag{16}$$

For $d, o \in \{k, c\}$, let $v^d_{j,(t)}$ be the values of score d on S_j sorted in *ascending* order. Sort recipient candidates in ascending order of $(\text{score } o, \text{id}_R)$ and assign

$$w^{d,o}_{R^o_{j,(t)}} = v^d_{j,(t)}. \tag{17}$$

The deterministic lexical ID is the zero-padded crop/query/component string `bbbb:qqqq:cccccc` (widths $4/4/6$); component indices start at 1. This is not the initial top-K tie rule: candidate selection uses descending confidence, descending area, ascending query index, then ascending component index. Area does not enter the factorial sorting key.

```
For each source j and each active-set study:
  S = original selected candidates
  If common-active: S = {R in S: min(kappa[R], C[R]) > EPS}
  For (d,o) in {(k,k),(c,c),(c,k),(k,c)}:
    recipients = sort S by (score[o], candidate_id), ascending
    values = sort score[d] on S, ascending
    Assign values to recipients in that order
    Omit assigned weights <= EPS; omit empty source groups
    Apply the image-normalized objective with these weights
```

Thus natural-active admission follows reassignment and can select different recipients across arms. Common-active filters the *raw paired scores before sorting*, then reassigns and normalizes on that restricted pool; it is not an intersection or post-hoc deletion of already-permuted natural outputs. The arms $(k,k), (c,c), (c,k), (k,c)$ are $D_kO_k, D_cO_c, D_cO_k, D_kO_c$. Excluded candidates receive no objective term, and active-source count is recomputed for each condition. All four arms, including D_kO_k, pay for C measurement on the original pool; common-active filtering does not eliminate that measurement cost.

Scope and numerical qualification. Every condition restores the original episode RNG and model/Adam reset state; the frozen experiment's optimizer, query readout and original-region loss locations remain. The archived OOM-retry iterator, old engine and old optimizer are not selected. The equations describe scalar allocation in real arithmetic. Actual weights are converted to the entropy tensor's native dtype, summed in retained term order, and normalized using the native objective; its denominator guard uses the sum when greater than 10^{-8} and 1 otherwise. Ideal float64 receipt coefficients are not substituted for native arithmetic. M04 separately records native-dtype weights, normalizers and coefficients. Image budget equalities are algebraic properties, not equal-gradient, equal-time, candidate-utility, or equivalence claims. This specification changes no result: the Object single-C versus legacy contrast remains negative, the PC60 contrast positive, the diagnostic-by-objective intervals cross zero, and the factorial effects remain conditional on distribution/order and active-set study. In M04, D identifies the multiset of score values and O the ordering assigned those values, with k for initial confidence and c for C. Arms are D_kO_k, D_cO_c, D_cO_k, and D_kO_c. Natural-active retains distribution-specific EPS admission; common-active conditions all four arms on their shared active candidates. The latter is a changed comparator, not unchanged production Confidence. The common-active schema name `native_C_minus_Confidence` denotes this conditioned $D_cO_c - D_kO_k$ contrast.

All four arms measure C. Object C distributions activate 171,161/173,503 candidates and PC60 101,002/101,505; common-active retains those same shared counts in every arm. All sources remain active. The complete 16 arm scores and 24 paired contrasts are shown in Tables 7 and 8. All arm and contrast intervals have zero undefined draws, with draws shared across both studies/all arms in each dataset.

Natural D_cO_c/D_kO_k points equal the separately executed native C/Confidence anchors, but their registered bootstrap namespaces differ; endpoint differences are not new samples or selected estimates. Object C ordering switches sign by distribution, while PC60 distribution replacement is null under C ordering and negative under Confidence ordering. Common-active shows the same qualitative pattern. Every arm pays 173,503/101,505 C diagnostic inputs and has 11,550/2,751 optimizer steps on Object/PC60, so this study cannot answer V01's zero-probe compute question.

The first CPU scorer attempt failed before annotation decoding because its validator used direct division rather than the pinned CUDA scalar reciprocal-multiply sequence. The failed frozen attempt, logs, and report remain; the replacement changed only unfrozen validation arithmetic and protocol binding, was independently reviewed/frozen, and succeeded without prediction reruns. Exact coefficient equality was not weakened by a tolerance. A successful execution is not a second scientific outcome selected over a negative one.

Table 7: M04: all natural-active and common-active factorial arms, pooled mIoU with pointwise 95% intervals. Common-active conditions on the shared active set, rather than silently replacing the natural comparator.

Dataset	Study	Arm / contrast	Estimate [95% CI]	U / draws
Object	Natural	Conf. dist. / Conf. order	29.361 [28.562, 30.015]	0/10000
Object	Natural	C dist. / C order	32.671 [31.785, 33.292]	0/10000
Object	Natural	C dist. / Conf. order	30.963 [30.172, 31.515]	0/10000
Object	Natural	Conf. dist. / C order	28.636 [27.783, 29.367]	0/10000
Object	Common-active	Conf. dist. / Conf. order	29.369 [28.562, 30.028]	0/10000
Object	Common-active	C dist. / C order	32.671 [31.785, 33.292]	0/10000
Object	Common-active	C dist. / Conf. order	30.911 [30.137, 31.462]	0/10000
Object	Common-active	Conf. dist. / C order	28.743 [27.901, 29.473]	0/10000
PC60	Natural	Conf. dist. / Conf. order	30.573 [30.086, 30.998]	0/10000
PC60	Natural	C dist. / C order	30.783 [30.277, 31.230]	0/10000
PC60	Natural	C dist. / Conf. order	30.142 [29.653, 30.575]	0/10000
PC60	Natural	Conf. dist. / C order	30.675 [30.195, 31.097]	0/10000
PC60	Common-active	Conf. dist. / Conf. order	30.584 [30.099, 31.009]	0/10000
PC60	Common-active	C dist. / C order	30.783 [30.277, 31.230]	0/10000
PC60	Common-active	C dist. / Conf. order	30.152 [29.664, 30.586]	0/10000
PC60	Common-active	Conf. dist. / C order	30.669 [30.188, 31.092]	0/10000

Table 8: M04: all six contrasts per dataset and active-set study, in mIoU percentage points. Interaction is the distribution-by-order difference-in-differences, not a candidate-causal effect. Negative and zero-crossing intervals remain.

Dataset	Study	Arm / contrast	Estimate [95% CI]	U / draws
Object	Natural	C minus Confidence	+3.310 [+2.656, +3.876]	0/10000
Object	Natural	Order @ C dist.	+1.708 [+1.224, +2.152]	0/10000
Object	Natural	Order @ Conf. dist.	-0.726 [-1.118, -0.308]	0/10000
Object	Natural	Dist. @ C order	+4.036 [+3.284, +4.659]	0/10000
Object	Natural	Dist. @ Conf. order	+1.602 [+0.985, +2.179]	0/10000
Object	Natural	Dist.-order interaction	+2.433 [+1.845, +2.957]	0/10000
Object	Common-active	C minus Confidence	+3.302 [+2.656, +3.854]	0/10000
Object	Common-active	Order @ C dist.	+1.760 [+1.268, +2.201]	0/10000
Object	Common-active	Order @ Conf. dist.	-0.626 [-0.952, -0.283]	0/10000
Object	Common-active	Dist. @ C order	+3.928 [+3.163, +4.552]	0/10000
Object	Common-active	Dist. @ Conf. order	+1.542 [+0.931, +2.113]	0/10000
Object	Common-active	Dist.-order interaction	+2.386 [+1.820, +2.865]	0/10000
PC60	Natural	C minus Confidence	+0.210 [+0.065, +0.348]	0/10000
PC60	Natural	Order @ C dist.	+0.641 [+0.483, +0.792]	0/10000
PC60	Natural	Order @ Conf. dist.	+0.102 [+0.057, +0.149]	0/10000
PC60	Natural	Dist. @ C order	+0.108 [-0.026, +0.240]	0/10000
PC60	Natural	Dist. @ Conf. order	-0.431 [-0.534, -0.330]	0/10000
PC60	Natural	Dist.-order interaction	+0.539 [+0.392, +0.683]	0/10000
PC60	Common-active	C minus Confidence	+0.199 [+0.057, +0.335]	0/10000
PC60	Common-active	Order @ C dist.	+0.631 [+0.471, +0.783]	0/10000
PC60	Common-active	Order @ Conf. dist.	+0.085 [+0.039, +0.130]	0/10000
PC60	Common-active	Dist. @ C order	+0.114 [-0.019, +0.242]	0/10000
PC60	Common-active	Dist. @ Conf. order	-0.432 [-0.533, -0.331]	0/10000
PC60	Common-active	Dist.-order interaction	+0.546 [+0.395, +0.689]	0/10000

Table 9: All 15 primary severity-5 corruption conditions: No Adapt, Confidence, and C pooled mIoU, with the registered C-minus-Confidence pointwise interval. Elastic transformation is negative; several other intervals cross zero. These are the original G-primary intervals, not replaced by the G03 follow-up namespace.

Condition	NoAdapt	Confidence	C	C − Conf. [95% CI]
Gaussian Noise	64.374	68.579	68.299	-0.280 [-1.605, +1.023]
Shot Noise	67.584	71.966	72.977	+1.011 [-0.109, +2.088]
Impulse Noise	64.397	69.499	68.513	-0.985 [-2.190, +0.195]
Defocus Blur	75.017	76.615	77.325	+0.710 [-0.426, +1.872]
Glass Blur	73.563	74.891	77.242	+2.351 [+1.278, +3.475]
Motion Blur	75.730	78.000	79.701	+1.701 [+0.523, +3.012]
Zoom Blur	60.080	65.890	66.859	+0.968 [-0.187, +2.076]
Snow	72.711	77.150	77.845	+0.695 [+0.015, +1.365]
Frost	66.406	70.646	72.006	+1.360 [+0.578, +2.147]
Fog	73.206	77.846	78.240	+0.393 [-0.541, +1.286]
Brightness	77.299	81.320	82.157	+0.837 [+0.208, +1.549]
Contrast	73.477	76.541	75.503	-1.038 [-2.269, +0.172]
Elastic Transform	64.411	70.548	69.542	-1.006 [-1.998, -0.046]
Pixelate	80.147	81.883	83.586	+1.703 [+0.839, +2.595]
Jpeg Compression	75.407	79.719	80.011	+0.292 [-0.404, +0.946]

Table 10: Predeclared G-primary macro and group summaries. Group estimates are equal-weight condition summaries under their original paired bootstrap, not additional selected corruption populations. Brightness belongs to the fixed weather group. Noise and digital intervals cross zero.

Condition	NoAdapt	Confidence	C	C − Conf. [95% CI]
Macro 15	70.921	74.739	75.320	+0.581 [+0.221, +0.947]
Noise	65.452	70.015	69.930	-0.085 [-0.908, +0.717]
Blur	71.097	73.849	75.282	+1.433 [+0.755, +2.101]
Weather	72.406	76.741	77.562	+0.821 [+0.367, +1.286]
Digital	73.361	77.172	77.160	-0.012 [-0.548, +0.529]

E Transfer and Fixed Corruptions

The G primary package contains 57 condition-arm cells, including NACLIP B/16 clean transfer, the declared SCLIP B/16 VOC20 condition, and all 15 L/14 VOC20 severity-5 corruptions. No Adapt is the same declared TRACE readout with zero updates. It is not a substituted published plain-NACLIP or SCLIP number. All corruption candidates, C values, and adaptations are constructed from corrupted inputs, without clean-image supervision.

The named SCLIP source is pinned to official commit `3608360267b6130c1ef18090d7289f17c771cb90`. A CPU official-method structural comparison and actual-model gates verify the tested component behavior; they do not establish that the last-9-layer fusion is the published whole SCLIP pipeline. The official CSA design (Wang, Mei, and Yuille 2024) is therefore attributed separately from the scope of the declared transfer test. Its negative -2.395064-pp C-minus-Confidence outcome is retained.

The fixed macro difference is $+0.580882$ pp; 11 condition point estimates are positive and four negative. Blur and weather group intervals are positive, while noise and digital cross zero. The all-15 table is not replaced by a favorable subset, and these pointwise intervals do not supply multiplicity-corrected discovery guarantees.

E.1 Five Severity Curves

G03 uses the five fixed types Gaussian noise, motion blur, fog, contrast, and JPEG compression, each at severities 1/3/5, and three arms No Adapt/Confidence/C. New severity-1/3 prediction has 30 cells and 43,470 evaluations; 15 severity-5 arm cells reuse exact original G prediction/count identities. Each condition has 1,449 ordered VOC20 sources grouped into 725 original batch-two episodes with a singleton tail. No severity-5 inference or whole huge per-image-count reread is performed.

The protocol SHA-256 is

`7655bee2573e1c79b8a4856f9272ce9b`
`45b5ccc025c1c09ca740f2d9b96cca86.`

It was registered after primary S5 results and before new S1/S3 predictions. Severity/type choices were already fixed and not selected by S5 outcomes; the follow-up is nevertheless not an untouched confirmatory evaluation. The scorer authenticates exact S5 source order/grouping, arm/readout/config identities, and compact original episode-count streams. It uses the original palette/OpenCV-nearest metric path, not native PIL. All 45 arm estimates and 15 contrasts appear in Tables 11 and 12.

The same 10,000 whole-original-episode draws are shared across these fixed curve arms under the G03 namespace. Pointwise intervals are not simultaneous bands. The new S5 endpoints can differ from primary-G endpoints without changing point evidence; neither is chosen as the preferred interval. No macro over the five types, 15-primary-condition replacement, AUC, fitted slope, severity winner, or monotonicity claim is added.

Table 11: All 45 G03 arm estimates: five fixed types, three severities, and three arms. Scores are pooled mIoU with pointwise 95% intervals. S5 arms reuse exact original evidence; they are not new model evaluations.

Corruption	Severity	Arm	mIoU [95% CI]	U / draws
Gaussian Noise	1	NoAdapt	79.102 [77.532, 80.467]	0/10000
Gaussian Noise	1	Confidence	81.420 [79.759, 82.775]	0/10000
Gaussian Noise	1	C	83.199 [81.636, 84.490]	0/10000
Gaussian Noise	3	NoAdapt	76.726 [75.109, 78.126]	0/10000
Gaussian Noise	3	Confidence	79.838 [78.225, 81.182]	0/10000
Gaussian Noise	3	C	81.713 [80.187, 82.999]	0/10000
Gaussian Noise	5	NoAdapt	64.374 [62.453, 66.120]	0/10000
Gaussian Noise	5	Confidence	68.579 [66.380, 70.556]	0/10000
Gaussian Noise	5	C	68.299 [66.100, 70.267]	0/10000
Motion Blur	1	NoAdapt	79.079 [77.535, 80.434]	0/10000
Motion Blur	1	Confidence	81.662 [80.084, 82.983]	0/10000
Motion Blur	1	C	83.144 [81.713, 84.349]	0/10000
Motion Blur	3	NoAdapt	79.041 [77.427, 80.458]	0/10000
Motion Blur	3	Confidence	80.168 [78.424, 81.631]	0/10000
Motion Blur	3	C	82.504 [80.878, 83.903]	0/10000
Motion Blur	5	NoAdapt	75.730 [73.799, 77.426]	0/10000
Motion Blur	5	Confidence	78.000 [75.952, 79.781]	0/10000
Motion Blur	5	C	79.701 [77.765, 81.369]	0/10000
Fog	1	NoAdapt	77.547 [75.998, 78.875]	0/10000
Fog	1	Confidence	80.456 [78.897, 81.746]	0/10000
Fog	1	C	82.097 [80.523, 83.428]	0/10000
Fog	3	NoAdapt	76.714 [75.129, 78.077]	0/10000
Fog	3	Confidence	79.997 [78.399, 81.321]	0/10000
Fog	3	C	80.806 [79.177, 82.216]	0/10000
Fog	5	NoAdapt	73.206 [71.501, 74.719]	0/10000
Fog	5	Confidence	77.846 [76.137, 79.280]	0/10000
Fog	5	C	78.240 [76.406, 79.862]	0/10000
Contrast	1	NoAdapt	78.926 [77.396, 80.234]	0/10000
Contrast	1	Confidence	81.701 [80.183, 82.993]	0/10000
Contrast	1	C	83.039 [81.553, 84.313]	0/10000
Contrast	3	NoAdapt	78.972 [77.361, 80.375]	0/10000
Contrast	3	Confidence	82.206 [80.665, 83.437]	0/10000
Contrast	3	C	83.007 [81.450, 84.336]	0/10000
Contrast	5	NoAdapt	73.477 [71.584, 75.174]	0/10000
Contrast	5	Confidence	76.541 [74.581, 78.250]	0/10000
Contrast	5	C	75.503 [73.297, 77.447]	0/10000
Jpeg Compression	1	NoAdapt	78.438 [76.900, 79.806]	0/10000
Jpeg Compression	1	Confidence	82.026 [80.456, 83.344]	0/10000
Jpeg Compression	1	C	83.438 [81.945, 84.676]	0/10000
Jpeg Compression	3	NoAdapt	78.322 [76.790, 79.607]	0/10000
Jpeg Compression	3	Confidence	82.379 [80.851, 83.662]	0/10000
Jpeg Compression	3	C	83.839 [82.378, 85.061]	0/10000
Jpeg Compression	5	NoAdapt	75.407 [73.762, 76.810]	0/10000
Jpeg Compression	5	Confidence	79.719 [77.999, 81.155]	0/10000
Jpeg Compression	5	C	80.011 [78.268, 81.526]	0/10000

Table 12: All 15 G03 C-minus-Confidence contrasts, mIoU percentage points. All ten S1/S3 intervals are positive; four S5 intervals cross zero, including two negative points. No null draw was discarded; undefined counts are zero.

Corruption	Severity	C − Conf. [95% CI]	U / draws
Gaussian Noise	1	+1.779 [+0.977, +2.600]	0/10000
Gaussian Noise	3	+1.875 [+0.969, +2.812]	0/10000
Gaussian Noise	5	-0.280 [-1.637, +1.032]	0/10000
Motion Blur	1	+1.482 [+0.862, +2.119]	0/10000
Motion Blur	3	+2.336 [+1.346, +3.350]	0/10000
Motion Blur	5	+1.701 [+0.532, +2.996]	0/10000
Fog	1	+1.641 [+1.034, +2.316]	0/10000
Fog	3	+0.809 [+0.014, +1.563]	0/10000
Fog	5	+0.393 [-0.550, +1.296]	0/10000
Contrast	1	+1.339 [+0.719, +2.008]	0/10000
Contrast	3	+0.802 [+0.093, +1.512]	0/10000
Contrast	5	-1.038 [-2.253, +0.179]	0/10000
Jpeg Compression	1	+1.412 [+0.745, +2.119]	0/10000
Jpeg Compression	3	+1.460 [+0.712, +2.221]	0/10000
Jpeg Compression	5	+0.292 [-0.407, +0.957]	0/10000

Table 13: All ten completed V01 grid points. Model seconds include episode zero and the common finite-value observer. Accuracy is pooled complete-population mIoU; no point is interpolated or selected by accuracy for matching.

Dataset	Arm	Steps	Model seconds	mIoU [95% CI]	U / draws
Object	C	15	974.522	32.575 [31.749, 33.165]	0/10000
Object	C	30	1626.462	32.671 [31.776, 33.293]	0/10000
Object	Confidence	30	1343.934	29.361 [28.550, 30.007]	0/10000
Object	Confidence	60	2653.890	25.841 [24.941, 26.624]	0/10000
Object	Confidence	120	5272.454	24.243 [23.399, 25.000]	0/10000
PC60	C	4	339.304	30.517 [30.002, 30.950]	0/10000
PC60	C	7	447.217	30.783 [30.259, 31.218]	0/10000
PC60	Confidence	7	281.004	30.573 [30.069, 30.986]	0/10000
PC60	Confidence	14	532.888	30.163 [29.652, 30.598]	0/10000
PC60	Confidence	28	1036.379	28.339 [27.804, 28.829]	0/10000

Table 14: V01 matching outcomes and declared descriptive contrasts. Nearest Confidence is selected by absolute log time ratio, with fewer steps breaking ties, not by accuracy. Both ratios fail the fixed $[0.9, 1.1]$ eligibility band; both near-time accuracy comparisons are null. Same-step and nearest-point contrasts do not replace those null outcomes.

Dataset	Comparison	Nearest Conf./C time	Estimate [95% CI]	U / draws
Object	Same-step C - Conf.	`n/a`	+3.310 [+2.659, +3.865]	0/10000
Object	Nearest-time descriptive	0.826	+3.310 [+2.659, +3.865]	0/10000
Object	Eligible near-time	0.826	`null`	`null`
PC60	Same-step C - Conf.	`n/a`	+0.210 [+0.062, +0.350]	0/10000
PC60	Nearest-time descriptive	1.192	+0.620 [+0.399, +0.831]	0/10000
PC60	Eligible near-time	1.192	`null`	`null`

F Actual Compute Allocation

V01 fixes two C step counts and three Confidence counts per dataset: Object C15/30 and Confidence30/60/120; PC60 C4/7 and Confidence7/14/28. All 10 cells and 50,525 evaluations complete with the same original candidates, input order, model/reset, and dataset settings. The actual first-input gates cover all budgets before the complete runs. The primary timer covers synchronized reset, candidates, requested diagnostics, updates, final reconstruction/readout, and the common finite-Adam observer. Episode zero is included. RGB decode/H2D, I/O, GT scoring, auxiliary gates/preflight, and final cleanup are separately excluded; no auxiliary intermediate readout is inserted into the primary curves.

Native C30 and C7 use 1,626.462316 and 447.216557 model seconds respectively. The nearest Confidence/C ratios are 0.826293 (Object Conf30) and 1.191566 (PC60 Conf14). Failing the prespecified 10% tolerance means the equal-time question remains unanswered by this grid, even though same-step intervals favor C. PC60's $+0.620207$-pp nearest-point difference is descriptive; it cannot be relabeled equal time. The complete grid remains fixed after outcomes, without new intermediate budgets or accuracy-based stopping.

For a nonempty standard episode with P encoder crops, S actual update steps, and n candidates, the nominal C schedule uses $(S+2)P+n$ encoder inputs, versus $(S+2)P$ for zero-probe controls. Actual counters govern skips and empty episodes. One C view versus two diagnostic views halves diagnostic inputs for identical candidates, not total computation or wall time. Calls depend on view-batch size. Historical eight-protocol timing omits episode zero as warmup, unlike V01; the old table is retained under that scope below and is not substituted into the new matching analysis.

New sealed summaries retain raw and warmup-adjusted counters/timings, source hashes, and resources. Retained systemd memory-peak accounting is not used as a scientific process-memory cost. Equal observed times in M04, which measures C in every arm, cannot establish zero-probe compute value.

Table 15: Model-side time per image (ms), excluding the first episode as warmup. Timing includes reset, candidate admission, diagnostic views, optimization, and final readout; it excludes data loading, disk writes, and offline scoring. Extra time is relative to Confidence under the same TRACE-selected configuration.

Dataset	Confidence (ms)	TRACE (ms)	Extra time
VOC20	153.46	165.99	+8.17%
VOC21	273.45	343.73	+25.70%
PC59	115.71	186.77	+61.41%
PC60	51.90	84.43	+62.68%
Object	261.11	320.36	+22.69%
Stuff	101.68	159.09	+56.45%
COCO-CD	23.98	30.56	+27.45%
Waterbirds	26.46	37.99	+43.58%

Table 16: M05 cached pixel-transition summaries. Rates use their declared initially-correct/incorrect/valid-pixel denominators; paired differences and intervals retain negative and null outcomes. These are full-strategy pixel events, not candidate utility.

Dataset	Arm / contrast	Endpoint	Estimate [95% CI]
Object	Uniform	Correct retention	83.932 [not reported]
Object	Uniform	Wrong correction	43.356 [not reported]
Object	Uniform	Correct-to-wrong / pre-correct	16.068 [not reported]
Object	Uniform	Correct-to-wrong / valid	10.456 [not reported]
Object	Uniform	Net correct change	4.688 [not reported]
Object	Confidence	Correct retention	85.919 [not reported]
Object	Confidence	Wrong correction	38.855 [not reported]
Object	Confidence	Correct-to-wrong / pre-correct	14.081 [not reported]
Object	Confidence	Correct-to-wrong / valid	9.163 [not reported]
Object	Confidence	Net correct change	4.408 [not reported]
Object	C	Correct retention	89.472 [not reported]
Object	C	Wrong correction	34.041 [not reported]
Object	C	Correct-to-wrong / pre-correct	10.528 [not reported]
Object	C	Correct-to-wrong / valid	6.851 [not reported]
Object	C	Net correct change	5.039 [not reported]
PC60	Uniform	Correct retention	92.495 [not reported]
PC60	Uniform	Wrong correction	12.672 [not reported]
PC60	Uniform	Correct-to-wrong / pre-correct	7.505 [not reported]
PC60	Uniform	Correct-to-wrong / valid	3.740 [not reported]
PC60	Uniform	Net correct change	2.616 [not reported]
PC60	Confidence	Correct retention	92.586 [not reported]
PC60	Confidence	Wrong correction	12.745 [not reported]
PC60	Confidence	Correct-to-wrong / pre-correct	7.414 [not reported]
PC60	Confidence	Correct-to-wrong / valid	3.695 [not reported]
PC60	Confidence	Net correct change	2.698 [not reported]
PC60	C	Correct retention	93.372 [not reported]
PC60	C	Wrong correction	13.028 [not reported]
PC60	C	Correct-to-wrong / pre-correct	6.628 [not reported]
PC60	C	Correct-to-wrong / valid	3.303 [not reported]
PC60	C	Net correct change	3.232 [not reported]
Object	C - Confidence	Correct retention	+3.553 [+2.949, +4.186]
Object	C - Confidence	Wrong correction	-4.815 [-5.941, -3.703]
Object	C - Confidence	Correct-to-wrong / pre-correct	-3.553 [-4.186, -2.949]
Object	C - Confidence	Correct-to-wrong / valid	-2.312 [-2.718, -1.921]
Object	C - Confidence	Net correct change	+0.630 [+0.100, +1.165]
PC60	C - Confidence	Correct retention	+0.786 [+0.643, +0.932]
PC60	C - Confidence	Wrong correction	+0.283 [+0.105, +0.459]
PC60	C - Confidence	Correct-to-wrong / pre-correct	-0.786 [-0.932, -0.643]
PC60	C - Confidence	Correct-to-wrong / valid	-0.392 [-0.464, -0.320]
PC60	C - Confidence	Net correct change	+0.534 [+0.398, +0.669]

G Cached Groups and Pixel Transitions

M05 uses existing sealed predictions to report complete-population pixel events, without further adaptation. It distinguishes retention of initially correct pixels, correction of initially incorrect pixels, and net correctness. Object C-minus-Confidence retention is $+3.552948$ pp, correction -4.814519 pp, and net correctness $+0.630416$ pp. PC60 differences are $+0.785984$, $+0.282761$, and $+0.533546$ pp. Thus Object correction declines even while retention and net correctness improve; the method does not improve every event rate. These event-rate estimands differ from pooled mIoU and from M01's isolated soft-support harm.

A01 joins cached transitions to initial candidate metadata using fixed pre-outcome groups: candidate count, exact comparable-pair ranking agreement, source mean confidence bins $[0, .25, .5, .75, 1]$, and source median area fractions $\leq .01$, $\leq .1$, or $> .1$. These are source groups, not mask-local correctness. All 5,000 Object sources reorder; PC60 has only two same-order and 5,103 reordered sources. The two-source interval is undefined in 1,315/10,000 draws. Neither dataset contains a natural singleton-candidate source group. Missing groups and null intervals are not repaired by dropping draws, merging bins, or inventing candidates.

Object's 80-class foreground mean improves $+3.355306$ pp while its background declines -0.310327 pp. PC60 thing/stuff class-subset differences are $+0.239624$ and $+0.182507$ pp. These are descriptive subsets under fixed taxonomy, not new algorithm roles. Candidate-mask-local correctness and region-utility estimands remain uncomputed and are not implied by the

Table 17: A01 Object cached source-group outcomes under original fixed bins. Empty groups and undefined bootstrap draws retain nulls. N is source count; U counts undefined draws. All source/candidate coverage and all five event metrics remain in structured evidence. Groups are not candidate-local causal attribution or pooled group mIoU.

Source group	N	Retention [CI]	Correction [CI]	Net change [CI]	U / draws
All sources	5000	+3.553 [+2.947, +4.183]	-4.815 [-5.949, -3.722]	+0.630 [+0.107, +1.167]	0/10000
No candidates	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
One candidate	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
Multiple candidates	5000	+3.553 [+2.947, +4.183]	-4.815 [-5.949, -3.722]	+0.630 [+0.107, +1.167]	0/10000
Order: none	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
Order: single	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
No order comparison	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
Same order	0	`null` [`not reported`]	`null` [`not reported`]	`null` [`not reported`]	`not reported`
Reordered	5000	+3.553 [+2.947, +4.183]	-4.815 [-5.949, -3.722]	+0.630 [+0.107, +1.167]	0/10000
Conf. < 0.25	2	-2.044 [`null`]	-23.609 [`null`]	-9.492 [`null`]	1392/10000
Conf. 0.25–0.50	228	+6.491 [+3.977, +9.183]	-5.822 [-10.096, -1.340]	+2.862 [+0.766, +5.040]	0/10000
Conf. 0.50–0.75	2245	+4.121 [+3.348, +4.903]	-4.972 [-6.377, -3.584]	+0.914 [+0.221, +1.623]	0/10000
Conf. ≥ 0.75	2525	+2.765 [+2.153, +3.401]	-4.583 [-5.732, -3.501]	+0.185 [-0.370, +0.733]	0/10000
Area $\leq .01$	4958	+3.560 [+2.949, +4.192]	-4.803 [-5.932, -3.716]	+0.634 [+0.109, +1.169]	0/10000
Area .01–.10	41	+2.185 [+0.238, +4.456]	-5.785 [-15.032, +1.636]	-0.132 [-3.272, +2.607]	0/10000
Area $> .10$	1	+27.879 [`null`]	-36.239 [`null`]	+12.283 [`null`]	3663/10000

completed cache analysis.

Table 18: A01 PC60 cached source-group outcomes, using the same definitions as Table 17. Only two sources have the same order; the 1,315 undefined draws and null interval remain.

Source group	N	Retention [CI]	Correction [CI]	Net change [CI]	U / draws
All sources	5105	+0.786 [+0.643, +0.928]	+0.283 [+0.103, +0.456]	+0.534 [+0.398, +0.666]	0/10000
No candidates	0	null [not reported]	null [not reported]	null [not reported]	not reported
One candidate	0	null [not reported]	null [not reported]	null [not reported]	not reported
Multiple candidates	5105	+0.786 [+0.643, +0.928]	+0.283 [+0.103, +0.456]	+0.534 [+0.398, +0.666]	0/10000
Order: none	0	null [not reported]	null [not reported]	null [not reported]	not reported
Order: single	0	null [not reported]	null [not reported]	null [not reported]	not reported
No order comparison	0	null [not reported]	null [not reported]	null [not reported]	not reported
Same order	2	-0.143 [null]	+0.724 [null]	-0.103 [null]	1315/10000
Reordered	5103	+0.787 [+0.644, +0.929]	+0.283 [+0.103, +0.456]	+0.534 [+0.398, +0.666]	0/10000
Conf. < 0.25	0	null [not reported]	null [not reported]	null [not reported]	not reported
Conf. 0.25–0.50	124	+1.279 [+0.545, +2.140]	+0.550 [-0.365, +1.687]	+1.040 [+0.353, +1.797]	0/10000
Conf. 0.50–0.75	1352	+0.885 [+0.621, +1.144]	+0.220 [-0.082, +0.524]	+0.579 [+0.358, +0.802]	0/10000
Conf. ≥ 0.75	3629	+0.720 [+0.549, +0.896]	+0.298 [+0.082, +0.503]	+0.499 [+0.336, +0.664]	0/10000
Area $\leq .01$	4882	+0.804 [+0.655, +0.951]	+0.269 [+0.087, +0.443]	+0.535 [+0.398, +0.670]	0/10000
Area .01–.10	220	+0.430 [+0.039, +0.848]	+0.603 [-0.128, +1.341]	+0.513 [+0.041, +0.987]	0/10000
Area $> .10$	3	-0.100 [null]	-1.004 [null]	-0.177 [null]	467/10000

Table 19: A01 class-subset means under the exact frozen taxonomy. These average class IoUs from pooled per-class counts; they are not binary-role IoUs or source-group means. No class-subset confidence interval is claimed. The structured evidence export retains the complete class mapping and positive-union coverage.

Dataset	Class subset	Classes	NoAdapt	Uniform	Confidence	C
Object	foreground	80	29.254	26.995	28.875	32.231
Object	background	1	61.042	68.905	68.219	67.909
Object	thing	80	29.254	26.995	28.875	32.231
Object	stuff	0	null	null	null	null
PC60	foreground	59	29.402	31.095	31.076	31.288
PC60	background	1	2.335	0.884	0.887	0.987
PC60	thing	30	37.274	39.453	39.372	39.612
PC60	stuff	29	21.258	22.450	22.494	22.677

H Fixed Actual Cases

M12 fixes original source indices 0 and 1 for Object and PC60 before case generation, after aggregate M02 results were known. It uses saved candidates/supports/predictions and exactly admitted CPU GT/RGB access for rendering, without model inference or new accuracy estimates. The four source IDs are COCO 139/285 and VOC 2009_001286/2009_001288. Their first candidate queries are tv, grass, sofa, and flower. Measured initial-confidence/C pairs are respectively 0.996743/0.523093, 0.987361/0.004221, 0.945441/0.888686, and 0.999443/0.214489. These pairs are not correctness labels or weights from a new experiment.

Case generation preceded no new scientific selection: the protocol was fixed before these images were rendered, but not before aggregate results existed. Tables 20 and 21 print every semantic class/color pair; the enclosed `M12_rendering_legends.json` retains all four authenticated rendering legends and measurement keys. These are projections of existing metadata, not new predictions or scores. Gray denotes ignored label 255. Green/magenta in the overlay panels denote geometric support, not their semantic meaning in GT/prediction panels. Readable vector keys above each unchanged composite repeat its panel order and measurements; the original raster annotations are not the sole key. The manuscript uses only derived composites and rendering metadata, not raw RGB, GT, or prediction arrays. Cases do not prove representativeness, calibrated risk, or causal allocation.

Original query: **tv**; initial confidence: 0.996743; C support: 0.523093.

Top left: RGB	Top: GT	Top: Confidence	Top right: Native C
Bottom left: original support	Bottom: translated support	Bottom: translated prediction	Bottom right: global prediction



Figure 1: Object index 0, COCO 139, original candidate query tv. The stored 960-by-632 composite contains all eight complete 224-pixel tiles and recorded measurements. Green/magenta indicate original/translated preservation support, not attention, a newly computed mask, or a displayed actual transformed input. Palette colors denote category indices and are interpreted within the dataset, not a shared universal vocabulary.

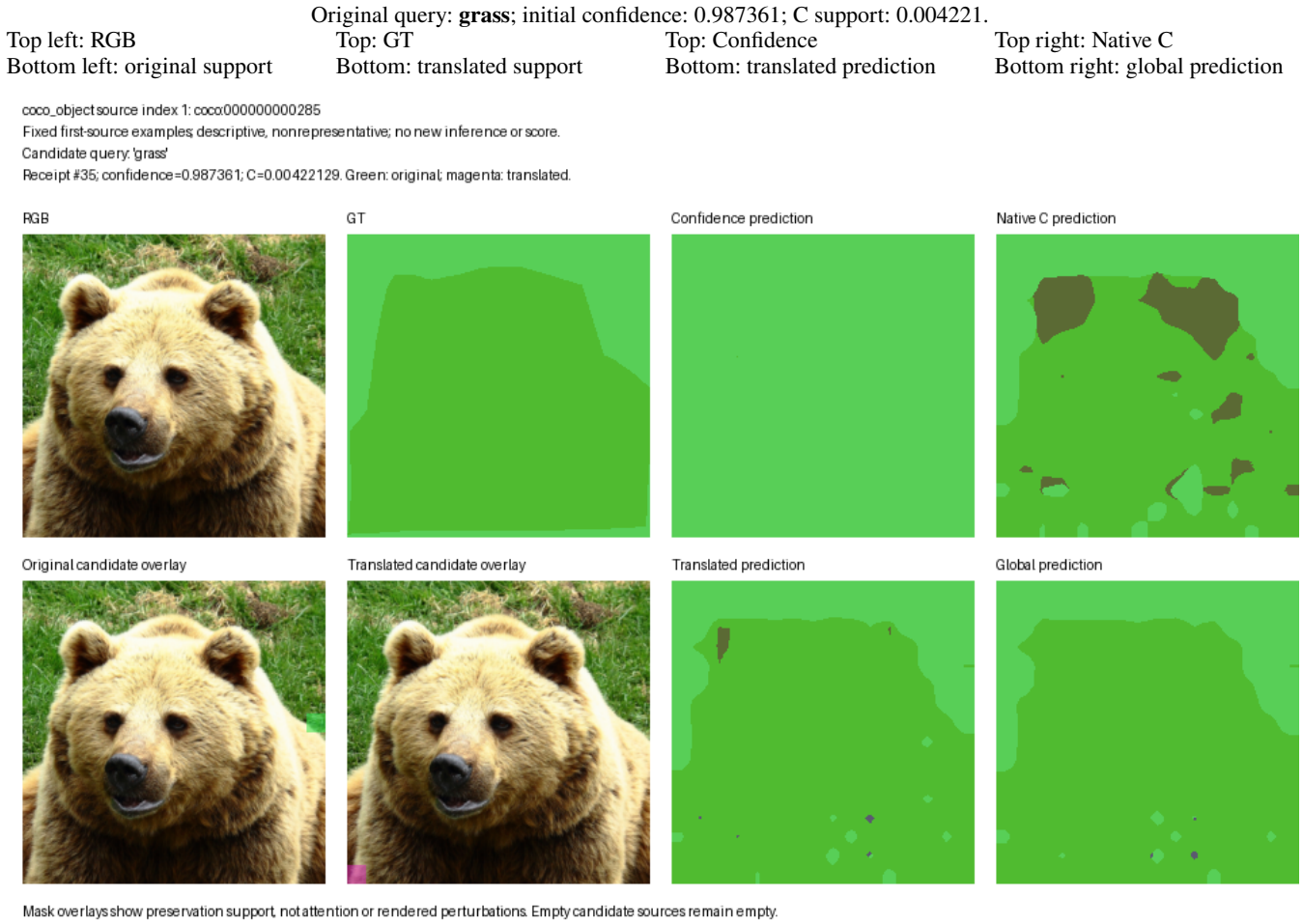


Figure 2: Object index 1, COCO 285, candidate query grass. This fixed case is retained although Confidence predicts almost background; its two palette colors mean it is not strictly a constant prediction. The display is descriptive, without a per-case winning criterion or prevalence claim.

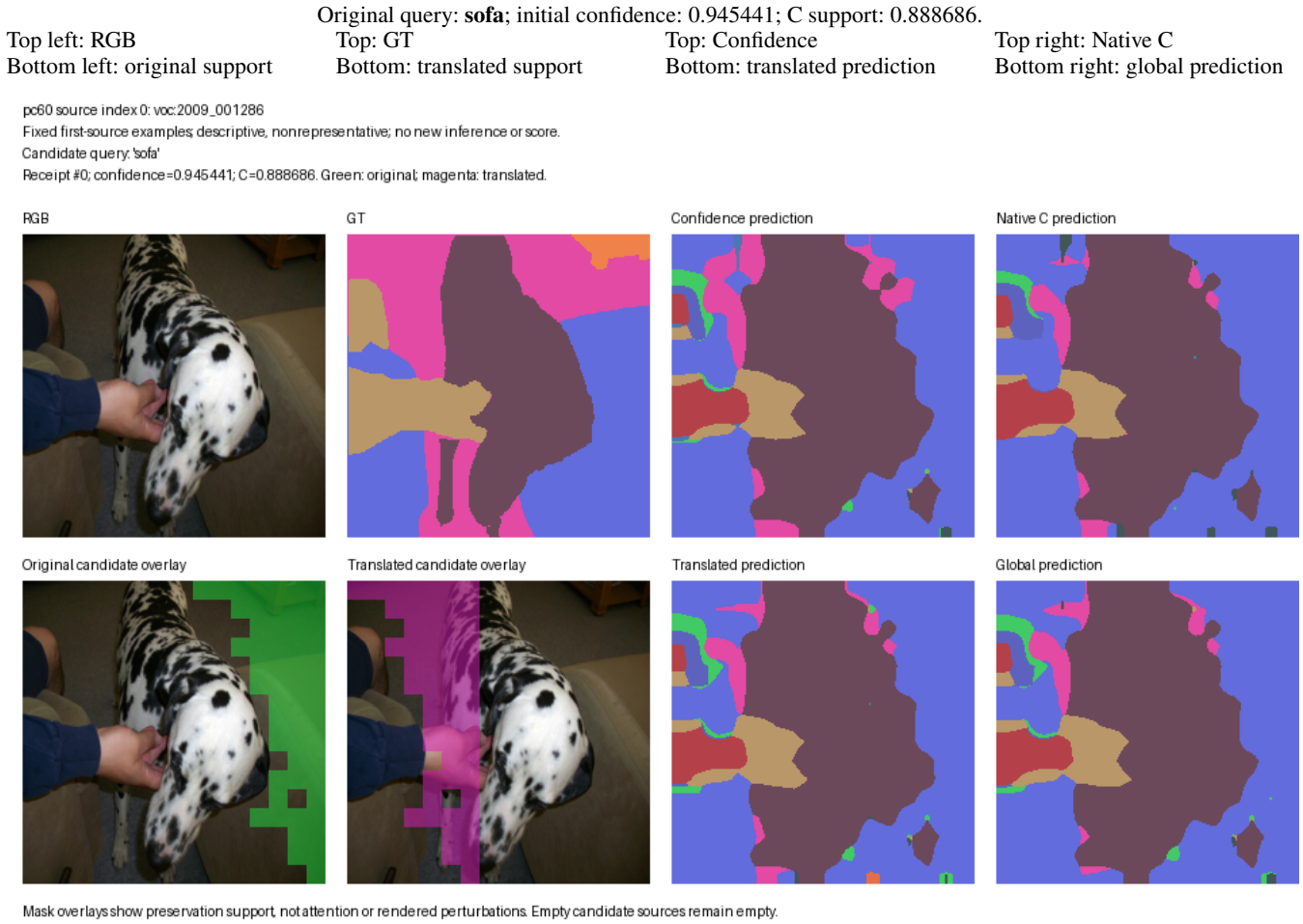


Figure 3: PC60 index 0, VOC 2009_001286, candidate query sofa. Original candidate identity and saved original-location support are retained. The four saved readouts belong to the specified pipeline controls, not external baselines. Neither the overlay nor a segmentation change identifies a causal effect of this candidate's weight under joint adaptation.

Original query: **flower**; initial confidence: 0.999443; C support: 0.214489.

Top left: RGB	Top: GT	Top: Confidence	Top right: Native C
Bottom left: original support	Bottom: translated support	Bottom: translated prediction	Bottom right: global prediction

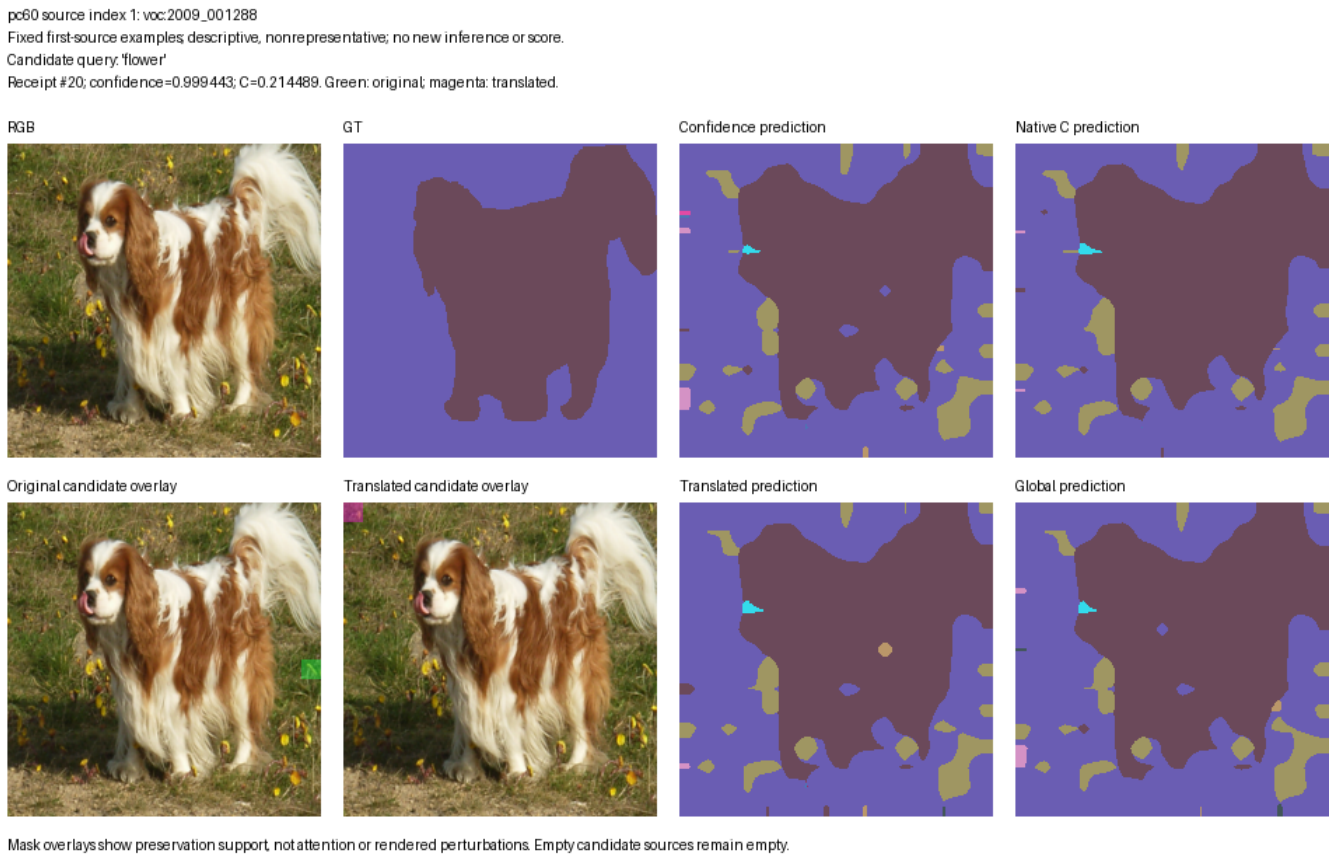


Figure 4: PC60 index 1, VOC 2009_001288, candidate query flower. All fixed examples remain, with no substitution based on ground truth, IoU improvement, or visual appeal. Ground truth entered only the separate admitted offline stage, not candidate construction or adaptation.

Table 20: Complete Object semantic palette copied from the authenticated M12 rendering metadata. ID is the evaluation category, not the candidate query index. RGB values are exact; ignore 255 is gray (128,128,128). These semantic colors are distinct from overlay roles.

ID	Color	Category	RGB	ID	Color	Category	RGB
0		background	88,207,87	41		wine glass	96,65,67
1		person	91,89,112	42		cup	151,53,87
2		bicycle	106,183,114	43		fork	212,146,197
3		car	88,109,215	44		knife	79,174,69
4		motorcycle	235,104,80	45		spoon	117,63,138
5		airplane	94,98,191	46		bowl	75,145,57
6		bus	55,208,197	47		banana	213,170,133
7		train	71,109,52	48		apple	159,94,182
8		truck	50,161,85	49		sandwich	98,108,220
9		boat	139,53,61	50		orange	238,130,74
10		traffic light	141,151,212	51		broccoli	95,144,183
11		fire hydrant	169,64,116	52		carrot	228,48,134
12		stop sign	82,231,60	53		hot dog	210,196,158
13		parking meter	50,109,106	54		pizza	152,208,107
14		bench	79,110,185	55		donut	77,50,88
15		bird	82,172,235	56		cake	174,186,60
16		cat	141,239,207	57		chair	76,68,107
17		dog	65,202,101	58		couch	82,155,65
18		horse	180,65,74	59		potted plant	77,118,180
19		sheep	188,74,170	60		bed	65,228,67
20		cow	57,133,171	61		dining table	221,97,134
21		elephant	61,204,237	62		toilet	82,133,201
22		bear	80,187,48	63		tv	86,86,101
23		zebra	107,73,90	64		laptop	191,134,143
24		giraffe	228,110,71	65		mouse	211,122,94
25		backpack	148,203,77	66		remote	56,122,127
26		umbrella	226,74,164	67		keyboard	73,231,63
27		handbag	159,150,98	68		cell phone	73,147,151
28		tie	51,219,236	69		microwave	87,209,97
29		suitcase	106,93,179	70		oven	71,124,217
30		frisbee	145,58,109	71		toaster	193,123,193
31		skis	227,157,212	72		sink	87,78,88
32		snowboard	224,102,58	73		refrigerator	176,155,137
33		sports ball	66,86,89	74		book	103,72,237
34		kite	88,79,112	75		clock	135,81,139
35		baseball bat	122,112,97	76		vase	96,91,58
36		baseball glove	127,161,169	77		scissors	155,78,227
37		skateboard	186,151,105	78		teddy bear	91,106,52
38		surfboard	211,59,162	79		hair drier	128,175,173
39		tennis racket	103,70,95	80		toothbrush	200,76,172
40		bottle	65,150,155				

Table 21: Complete PC60 semantic palette copied from the authenticated M12 rendering metadata. ID is the evaluation category, not the candidate query index. RGB values are exact; ignore 255 is gray (128,128,128). These semantic colors are distinct from overlay roles.

ID	Color	Category	RGB	ID	Color	Category	RGB
0		background	88,207,87	30		ground	145,58,109
1		aeroplane	91,89,112	31		horse	227,157,212
2		bag	106,183,114	32		keyboard	224,102,58
3		bed	88,109,215	33		light	66,86,89
4		bedclothes	235,104,80	34		motorbike	88,79,112
5		bench	94,98,191	35		mountain	122,112,97
6		bicycle	55,208,197	36		mouse	127,161,169
7		bird	71,109,52	37		person	186,151,105
8		boat	50,161,85	38		plate	211,59,162
9		book	139,53,61	39		platform	103,70,95
10		bottle	141,151,212	40		pottedplant	65,150,155
11		building	169,64,116	41		road	96,65,67
12		bus	82,231,60	42		rock	151,53,87
13		cabinet	50,109,106	43		sheep	212,146,197
14		car	79,110,185	44		shelves	79,174,69
15		cat	82,172,235	45		sidewalk	117,63,138
16		ceiling	141,239,207	46		sign	75,145,57
17		chair	65,202,101	47		sky	213,170,133
18		cloth	180,65,74	48		snow	159,94,182
19		computer	188,74,170	49		sofa	98,108,220
20		cow	57,133,171	50		table	238,130,74
21		cup	61,204,237	51		track	95,144,183
22		curtain	80,187,48	52		train	228,48,134
23		dog	107,73,90	53		tree	210,196,158
24		door	228,110,71	54		truck	152,208,107
25		fence	148,203,77	55		tvmonitor	77,50,88
26		floor	226,74,164	56		wall	174,186,60
27		flower	159,150,98	57		water	76,68,107
28		food	51,219,236	58		window	82,155,65
29		grass	106,93,179	59		wood	77,118,180

Table 22: Published references (R) and local TRACE (L), mIoU (%). Reported means and standard deviations follow Table 5, Original NACLIP, of Noori et al. (2025); No Adapt has no reported dispersion. Local results use seed 0 and TRACE-selected configurations on development-exposed populations. Published and local methods retain their respective protocols.

Method	VOC20	VOC21	PC60	Object	Stuff
No Adapt (R)	75.91	45.12	24.95	23.80	18.34
TENT (R)	77.00 ± 0.04	45.65 ± 0.02	25.29 ± 0.01	24.88 ± 0.01	18.76 ± 0.01
TPT (R)	75.93 ± 0.01	45.17 ± 0.01	24.98 ± 0.01	23.84 ± 0.01	18.35 ± 0.01
WATT (R)	57.73 ± 0.06	28.58 ± 0.05	14.77 ± 0.03	14.14 ± 0.06	9.49 ± 0.02
CLIPArTT (R)	72.77 ± 0.14	39.50 ± 0.04	21.88 ± 0.03	21.34 ± 0.03	15.48 ± 0.01
MLMP (R)	83.76 ± 0.00	50.78 ± 0.02	27.99 ± 0.03	28.84 ± 0.01	21.25 ± 0.01
TRACE (L)	83.61	53.38	30.78	32.67	21.91

Table 23: COCO-CD mIoU (%) alongside existing local native baselines. TENT is the strongest historical local baseline (72.20). Historical baselines and tuned TRACE retain their respective backends, prompts, and optimization settings; the historical No Adapt uses its native readout.

Method / protocol	COCO-CD
No Adapt (L, historical native)	61.14
TENT (L, historical native)	72.20
TPT (L, historical native)	64.27
MLMP (L, historical native)	71.01
TRACE (L, tuned)	74.23

I Published and Historical References

Published values (R) and local measurements (L) have different evidential roles. MLMP's Original NACLIP Table 5 reports its own protocols and dispersion (Noori et al. 2025). Local TRACE at the fixed selected settings exceeds four of its five listed MLMP point values, while VOC20 is lower by 0.145655 pp. Shared backbone family does not certify identical checkpoints, query vocabularies, preprocessing, readouts, budgets, or tuning histories. Table 22 is therefore a reference of operating points, not equal-protocol superiority.

COCO-CD historical native baselines use their original method backends/prompts and 10 updates at learning rate 0.001; the strongest is TENT at 72.200299 versus TRACE 74.231207. Episode composition and settings differ, so the 2.030908-pp gap is not an equal-budget attribution. Current matched No Adapt, not historical native No Adapt, defines the current adaptation gain. These are already completed references, not new baseline replications. No local WATT/CLIPArTT or restored baseline sweep is claimed.

J Provenance and Interpretation Checklist

The new paper exporter reads only explicitly allowlisted bounded metadata summaries, their exact global seals, and already verified terminal receipts. It checks known hashes before parsing, declared seal membership, and successful terminal bindings. No GT, raw RGB, model, prediction-array, huge per-image-count, or bootstrap resampling is read or run during paper export. Plot markers, error bars, and table values project precomputed estimates; structured JSON retains full precision, nulls, denominator coverage, bootstrap scope, and source hashes. Historical source-group cache analyses retain their distinct seal-only completion protocol rather than an invented service receipt.

All new fixed necessary experimental computation and offline scoring has completed, but that fact does not turn the study into independent unseen-data validation. Each scientific comparison retains its own fixed protocol, historical exposure, metric geometry, and bootstrap namespace. Negative Object-legacy/SCLIP/elastic outcomes, severe-corruption zero-crossing intervals, M01 sparse matching, and V01 unmatched outcomes remain part of the principal interpretation. No outcome selects a new production default, rerun, expanded budget grid, different corruption subset, or optional resource-conditional experiment.

Author-side checks ask five questions. Contribution: is the proposed object the specified allocation rule rather than an unsupported independent risk estimator? Clarity: are source/crop/token/candidate and query/category distinguished? Experimental strength: are fixed operating points separated from independent tuning and uncertainty? Completeness: are all signed primary, factorial, legacy, transfer, severe-corruption and compute results retained? Soundness: are normalization cancellation, EPS support, shared errors and joint Adam interactions explicit? The answers and source map are maintained in the companion claim ledger; fresh independent review remains separate from this author checklist.

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